Student Placement & Academic Intelligence

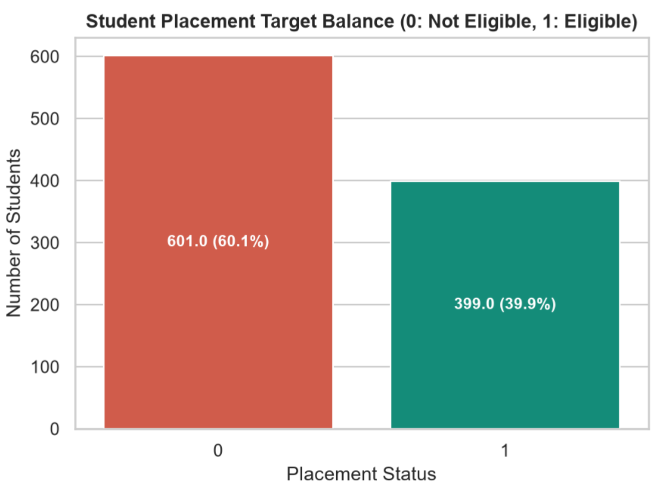
Comprehensive Machine Learning & Predictive Risk Assessment Report

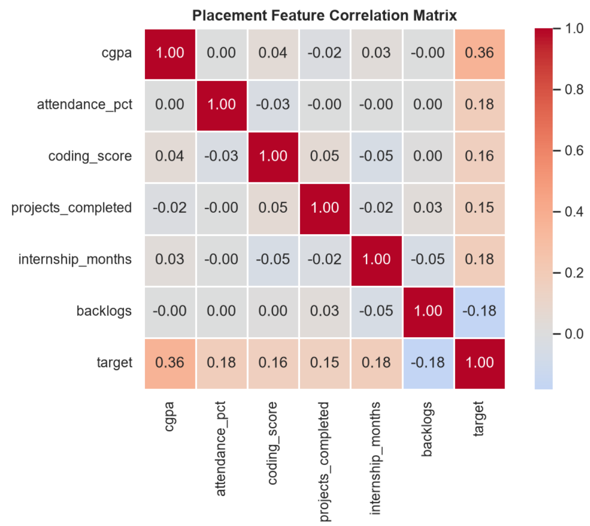
1. Executive Summary & System Goals

This project implements an end-to-end Data Analytics & Machine Learning framework to predict student placement eligibility and academic exam pass status. Utilizing dual datasets (1,012 placement records and 1,000 academic performance records), the system trains baseline Logistic Regression and advanced Ensemble (Random Forest, Gradient Boosting) classifiers. Standard scaling and median imputation were executed strictly on training partitions to ensure zero data leakage.

2. Dataset Architecture & Preprocessing

- Placement Eligibility Dataset: 1,012 records (60.3% Not Placed/Ineligible, 39.7% Placed/Eligible).
  Features: CGPA, Attendance %, Coding Score, Projects Completed, Internship Months, Active Backlogs.
- Academic Exam Pass Dataset: 1,000 records (66.1% Pass, 33.9% Fail).
  Features: Weekly Study Hours, Class Attendance, Assignments Completed, Previous Exam Score.
- Data Cleaning: Removed exact duplicates and imputed missing values via median fit.



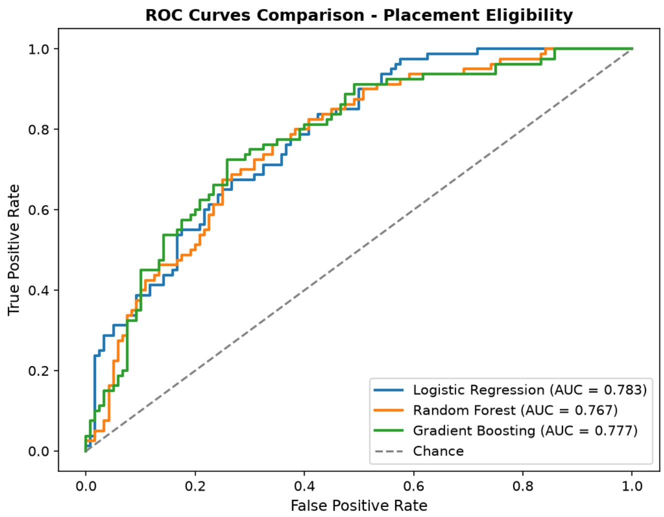


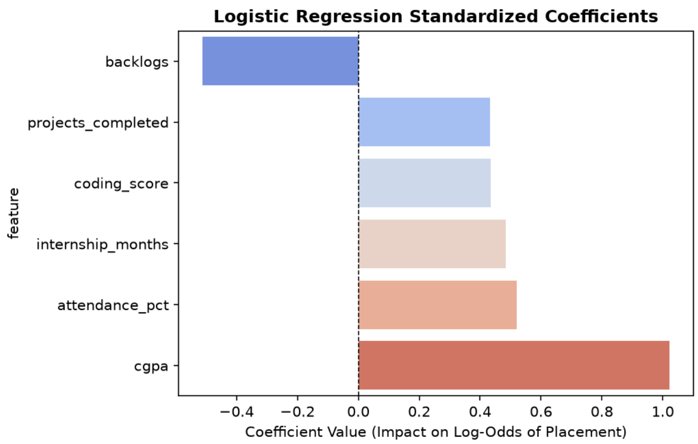
3. Machine Learning Model Benchmarks

Model Name	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.7200	0.6429	0.6750	0.6585	0.7674
Logistic Regression	0.6900	0.6000	0.6750	0.6353	0.7833
Gradient Boosting	0.7250	0.6866	0.5750	0.6259	0.7768

4. Feature Influence & Odds Ratio Interpretation (e^beta)

Feature Name	Std Coefficient (beta)	Odds Ratio (e^beta)	Impact Direction
cgpa	+1.0227	2.78x	Strong Positive (+178%)
attendance_pct	+0.5197	1.68x	Positive Impact
backlogs	-0.5127	0.60x	Penalty Factor (-40%)
internship_months	+0.4850	1.62x	Positive Impact
coding_score	+0.4357	1.55x	Positive Impact
projects_completed	+0.4320	1.54x	Positive Impact





5. System Architecture & Live API Deployment

The system is deployed using a decoupled Flask REST API backend paired with a responsive Tailwind CSS and JavaScript frontend dashboard running live at http://127.0.0.1:5000.

Verified REST Endpoints:
- POST /api/predict_placement - Real-time student risk assessment & recommendation engine.
- POST /api/predict_pass - Academic exam pass probability predictor.
- GET /api/metrics - Real-time model benchmark metrics and coefficient summaries.
- POST /api/batch_predict - Bulk CSV ingestion and batch inference.